Trade and Labor Market Dynamics: Rewriting as a Mean-Field Game

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1 Trade and Labor Market Dynamics Summary

Households have Cobb-Douglas preferences

$$C_t^{nj} = \prod_{k=1}^{J} \left(c_t^{nj,k}\right)^{\alpha^k}, \sum_{k=1}^{J} \alpha^k = 1$$

where $c_t^{nj,k}$ is the consumption of sector k goods in market nj at time t. The ideal price index is given by

$$P_t^n = \prod_{k=1}^{J} \left(\frac{P_t^{nk}}{\alpha^k}\right)^{\alpha^k}$$

where P_t^{nk} is the price index of goods purchased from sector k for final consumption in region n. Households are forward looking and discount the future rate $\beta \geq 0$. Migration decisions are subject to sectoral and spatial mobility costs. Labor relocation costs are given by $\tau^{nj,ik} \geq 0$ depend on the origin (nj) and destination (ik) are time invariant, additive, and measured in terms of utility. Households also have an additive idiosyncratic shocks for each choice, denoted by ε_t^{ik}.

At each period, households observe the conditions in all labor markets and the realizations of their own idiosyncratic shocks. If they begin the period in a labor market, they work and earn the market wage. At the end of the period, households have the option to relocate. Household decision to relocate is governed by

$$v_t^{nj} = U\left(C_t^{nj}\right) + \max_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta E\left[v_{t+1}^{ik}\right] - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\}, \tag{1}$$

$$\text{s.t. } C_t^{nj} = \frac{w_t^{nj}}{P_t^n}$$

where v_t^{nj} is the lifetime utility of a household currently in region n and sector j at time t. We assume that idiosyncratic shocks are i.i.d. over time and distributed Type-1 Extreme Value with mean zero and ν controls the variance of these shocks.

The expected lifetime utility of a representative agent in market nj at time t is thus given by

$$V_t^{nj} \equiv E\left[v_t^{nj}\right]$$

and using the assumption of the distribution of the shocks, we can rewrite the expected lifetime utility as

$$V_t^{nj} = U\left(C_t^{nj}\right) + \nu \log\left(\beta \sum_{ik} \exp\left(\beta V_{t+1}^{ik} - \tau^{nj,ik}\right)^{1/\nu}\right) \tag{2}$$

(proof in the Appendix). which is the Bellman equation of our dynamic programming (DP) problem. In a mean-field game, this can be thought of us as the backward equation. This process describes how decisions for the future are made.

The transition matrix governing the probabilities that agents relocate from market nj to ik is determined by

$$\mu_t^{nj,ik} = \frac{\exp\left(\beta V_{t+1}^{ik} - \tau^{nj,ik}\right)^{1/\nu}}{\sum_{mh} \exp\left(\beta V_{t+1}^{mh} - \tau^{nj,mh}\right)^{1/\nu}}$$

The transition matrix follows a gravity structure and the parameter $\frac{1}{\nu}$ can be interpreted as the elasticity of migration. The final equation that characterizes our DP problem is given by

$$L_{t+1}^{nj} = \sum_{ik} \mu_t^{ik,nj} L_t^{ik}$$

which describes the evolution of labor markets. Within the context of mean-field game, this would be the Kolmogorov forward equation.

2 Appendix

Deriving equation (2).

$$v_t^{nj} = U\left(C_t^{nj}\right) + \max_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta E\left[v_{t+1}^{ik}\right] - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\}$$

$$E\left[v_t^{nj}\right] = E\left[U\left(C_t^{nj}\right) + \max_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta E\left[v_{t+1}^{ik}\right] - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\}\right]$$

$$V_t^{nj} = U\left(C_t^{nj}\right) + E\left[\max_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\}\right]$$

We are concerned with deriving a closed form solution for this term

$$\Phi_t^{nj} = E\left[\max_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\}\right]$$

When agents decide to relocate from market nj to ik, they need to choose a location that maximizes their expected lifetime utility. Suppose the solutions that maximizes our agent's expected lifetime utility is given by

$$\text{argmax}_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\} = ik$$

$$\max_{\{i,k\}_{i=1,k=0}^{N,J}} \left\{\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right\} = \beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}$$

This can only be true if for all $mh \neq ik$, $\beta V_{t+1}^{mh} - \tau^{nj,mh} + \nu\varepsilon_t^{mh} \leq \beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}$. Rewriting this expression, we can say

$$\varepsilon_t^{mh} - \varepsilon_t^{ik} \leq \frac{\beta\left(V_{t+1}^{ik} - V_{t+1}^{mh}\right) - \left(\tau^{nj,ik} - \tau^{nj,mh}\right)}{\nu}$$

let $\bar{\varepsilon}_t^{ik,mh} = \frac{\beta\left(V_{t+1}^{ik} - V_{t+1}^{mh}\right) - \left(\tau^{nj,ik} - \tau^{nj,mh}\right)}{\nu}$, which implies

$$\varepsilon_t^{mh} \leq \bar{\varepsilon}_t^{ik,mh} + \varepsilon_t^{ik}$$

Therefore, the probability that the idiosyncratic shock for every $mh \neq ik$ does not maximize expected future utility is therefore given by the joint probability density $f\left(\varepsilon_t^{ik}\right) \prod_{mh \neq ik} F\left(\bar{\varepsilon}_t^{ik,mh} + \varepsilon_t^{ik}\right)$. Taking the expectation over the continuum of idiosyncratic shocks thus takes the form

$$\int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu\varepsilon_t^{ik}\right) f\left(\varepsilon_t^{ik}\right) \prod_{mh \neq ik} F\left(\bar{\varepsilon}_t^{ik,mh} + \varepsilon_t^{ik}\right) d\varepsilon_t^{ik}$$

This gives the contribution from a single market. Therefore, by the law of total expectation, the average option value from moving from market nj can be found by summing over all households

$$\Phi_t^{nj} = \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \varepsilon_t^{ik}\right) f\left(\varepsilon_t^{ik}\right) \prod_{mh \neq ik} F\left(\bar{\varepsilon}_t^{ik,mh} + \varepsilon_t^{ik}\right) d\varepsilon_t^{ik}$$

We assume that the idiosyncratic shocks follow a Type-1 Extreme Value distribution. Substituting the PDF and CDF and with some algebra, we get

$$\begin{aligned}
\Phi_t^{nj} &= \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \varepsilon_t^{ik}\right) f\left(\varepsilon_t^{ik}\right) \prod_{mh \neq ik} F\left(\bar{\varepsilon}_t^{ik,mh} + \varepsilon_t^{ik}\right) d\varepsilon_t^{ik} \\
&= \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \varepsilon_t^{ik}\right) e^{\left(-\varepsilon_t^{ik} - \gamma\right)} e^{\left(-e^{\left(-\varepsilon_t^{ik} - \gamma\right)} \sum_{ik} e^{\left(-\bar{\varepsilon}_t^{ik,mh}\right)}\right)} d\varepsilon_t^{ik} \\
\lambda_t^{ik} &\equiv \log\left(\sum_{mh} e^{\left(-\bar{\varepsilon}_t^{ik,mh}\right)}\right) \\
\zeta_t^{ik} &= \varepsilon_t^{ik} + \gamma \\
\gamma &\equiv \text{Euler's constant} \\
d\zeta_t^{ik} &= d\varepsilon_t^{ik} \\
\Phi_t^{nj} &= \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \varepsilon_t^{ik}\right) e^{\left(-\zeta_t^{ik}\right)} e^{\left(-e^{\left(-\zeta_t^{ik}\right)} e^{\lambda_t^{ik}}\right)} d\varepsilon_t^{ik} \\
&= \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \left(\zeta_t^{ik} - \gamma\right)\right) \exp\left(-\zeta_t^{ik} - \exp\left(-\left(\zeta_t^{ik} - \lambda_t^{ik}\right)\right)\right) d\zeta_t^{ik} \\
\tilde{y}_t^{ik} &= \zeta_t^{ik} - \lambda_t^{ik} \\
d\tilde{y}_t^{ik} &= d\zeta_t^{ik} \\
\Phi_t^{nj} &= \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \left(\tilde{y}_t^{ik} + \lambda_t^{ik} - \gamma\right)\right) \exp\left(-\left(\lambda_t^{ik} + \tilde{y}_t^{ik}\right) - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\zeta_t^{ik} \\
\Phi_t^{nj} &= \sum_{ik} \int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} + \nu \tilde{y}_t^{ik} + \nu \lambda_t^{ik} - \nu\gamma\right) \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\zeta_t^{ik} \\
\Phi_t^{nj} &= \sum_{ik} \left[\int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} - \nu\gamma\right) \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\zeta_t^{ik}\right] \\
&\quad + \sum_{ik} \left[\int_{-\infty}^{\infty} \nu \tilde{y}_t^{ik} \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\zeta_t^{ik}\right] \\
&\quad + \sum_{ik} \left[\int_{-\infty}^{\infty} \nu \lambda_t^{ik} \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\zeta_t^{ik}\right] \\
\Phi_t^{nj} &= \sum_{ik} \left[\int_{-\infty}^{\infty} \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} - \nu\gamma\right) \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\tilde{y}_t^{ik}\right] \\
&\quad + \sum_{ik} \left[\int_{-\infty}^{\infty} \nu \tilde{y}_t^{ik} \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\tilde{y}_t^{ik}\right] \\
&\quad + \sum_{ik} \left[\int_{-\infty}^{\infty} \nu \lambda_t^{ik} \exp\left(-\lambda_t^{ik}\right) \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\tilde{y}_t^{ik}\right] \\
\Phi_t^{nj} &= \sum_{ik} \left[\exp\left(-\lambda_t^{ik}\right) \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} - \nu\gamma\right) \int_{-\infty}^{\infty} \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\tilde{y}_t^{ik}\right] \\
&\quad + \sum_{ik} \left[\nu \exp\left(-\lambda_t^{ik}\right) \int_{-\infty}^{\infty} \tilde{y}_t^{ik} \exp\left(-\tilde{y}_t^{ik} - \exp\left(-\tilde{y}_t^{ik}\right)\right) d\tilde{y}_t^{ik}\right]
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{ik}\left[\nu\lambda_t^{ik}\exp\left(-\lambda_t^{ik}\right)\int_{-\infty}^{\infty}\exp\left(-\tilde{y}_t^{ik}-\exp\left(-\tilde{y}_t^{ik}\right)\right)d\tilde{y}_t^{ik}\right]\\
\Phi_t^{nj} &= \sum_{ik}\left[\exp\left(-\lambda_t^{ik}\right)\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}-\nu\gamma\right)\int_{-\infty}^{\infty}\exp\left(-\tilde{y}_t^{ik}-\exp\left(-\tilde{y}_t^{ik}\right)\right)d\tilde{y}_t^{ik}\right]\\
&+\sum_{ik}\left[\nu\gamma\exp\left(-\lambda_t^{ik}\right)\right]\\
&+\sum_{ik}\left[\nu\lambda_t^{ik}\exp\left(-\lambda_t^{ik}\right)\int_{-\infty}^{\infty}\exp\left(-\tilde{y}_t^{ik}-\exp\left(-\tilde{y}_t^{ik}\right)\right)d\tilde{y}_t^{ik}\right]
\end{aligned}
$$

$$
\int_{-\infty}^{\infty}\exp\left(-x-\exp\left(-x\right)\right)dx = 1
$$

$$
\begin{aligned}
\Phi_t^{nj} &= \sum_{ik}\exp\left(-\lambda_t^{ik}\right)\left[\beta V_{t+1}^{ik}-\tau^{nj,ik}+\nu\lambda_t^{ik}\right]\\
\Phi_t^{nj} &= \sum_{ik}\left[\exp\left(-\log\left(\sum_{mh}\exp\left(-\bar{\varepsilon}_t^{ik,mh}\right)\right)\right)\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)\right]\\
&+\sum_{ik}\left[\exp\left(-\log\left(\sum_{mh}\exp\left(-\bar{\varepsilon}_t^{ik,mh}\right)\right)\right)\nu\log\left(\sum_{mh}\exp\left(-\bar{\varepsilon}_t^{ik,mh}\right)\right)\right]\\
\bar{\varepsilon}_t^{ik,mh} &= \frac{\beta\left(V_{t+1}^{ik}-V_{t+1}^{mh}\right)-\left(\tau^{nj,ik}-\tau^{nj,mh}\right)}{\nu}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_t^{nj} &= \sum_{ik}\left[\exp\left(-\log\left(\sum_{mh}\exp\left(-\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)\frac{1}{\nu}\right)\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)\frac{1}{\nu}\right)\right)\right)\left(\begin{array}{c}\beta V_{t+1}^{ik}-\tau^{nj,ik}\\+\nu\log\left(\sum_{mh}\exp\left(-\bar{\varepsilon}_t^{ik,mh}\right)\right)\end{array}\right)\right]\\
\Phi_t^{nj} &= \sum_{ik}\left[\exp\left(-\log\left(\frac{\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)\frac{1}{\nu}\right)}{\exp\left(\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)\frac{1}{\nu}\right)}\right)\right)\left(\begin{array}{c}\beta V_{t+1}^{ik}-\tau^{nj,ik}\\+\nu\log\left(\sum_{mh}\exp\left(-\bar{\varepsilon}_t^{ik,mh}\right)\right)\end{array}\right)\right]\\
\Phi_t^{nj} &= \sum_{ik}\left[\frac{\exp\left(\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)^{\frac{1}{\nu}}\right)}{\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)^{\frac{1}{\nu}}\right)}\left(\begin{array}{c}\beta V_{t+1}^{ik}-\tau^{nj,ik}\\+\nu\log\left(\sum_{mh}\exp\left(-\bar{\varepsilon}_t^{ik,mh}\right)\right)\end{array}\right)\right]\\
\Phi_t^{nj} &= \sum_{ik}\left[\frac{\exp\left(\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)^{\frac{1}{\nu}}\right)}{\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)^{\frac{1}{\nu}}\right)}\left(\begin{array}{c}\beta V_{t+1}^{ik}-\tau^{nj,ik}\\+\nu\log\left(\frac{\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)^{\frac{1}{\nu}}\right)}{\exp\left(\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)^{\frac{1}{\nu}}\right)}\right)\end{array}\right)\right]\\
\Phi_t^{nj} &= \sum_{ik}\left[\frac{\exp\left(\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)^{\frac{1}{\nu}}\right)}{\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)^{\frac{1}{\nu}}\right)}\left(\begin{array}{c}\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)-\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)\\+\nu\log\left[\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)\frac{1}{\nu}\right)\right]\end{array}\right)\right]\\
\Phi_t^{nj} &= \left[\frac{\sum_{ik}\exp\left(\left(\beta V_{t+1}^{ik}-\tau^{nj,ik}\right)^{\frac{1}{\nu}}\right)}{\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)^{\frac{1}{\nu}}\right)}\left(\nu\log\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)\right)^{\frac{1}{\nu}}\right)\right]\\
\Phi_t^{nj} &= \nu\log\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)\right)^{\frac{1}{\nu}}
\end{aligned}
$$

Substituting this term back into equation (2), we arrive at our final result

$$
\begin{aligned}
V_t^{nj} &= U\left(C_t^{nj}\right)+E\left[\max_{\{i,k\}_{i=1,k=0}^{N,J}}\left\{\beta V_{t+1}^{ik}-\tau^{nj,ik}+\nu\varepsilon_t^{ik}\right\}\right]\\
V_t^{nj} &= U\left(C_t^{nj}\right)+\nu\log\sum_{mh}\exp\left(\left(\beta V_{t+1}^{mh}-\tau^{nj,mh}\right)\right)^{\frac{1}{\nu}}
\end{aligned}
$$

3 Sources

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